Problem Chosen	2025 HiMCM Summary Sheet	Team Control Number
A		16486

Building sweep is the process that responders verifying each area in a building is clear of occupants during emergencies [1] such as fires or gas leaks. This process is complex due to the variation of buildings' size, layout, occupants' distribution. Smoke, blocked corridors, and limited communication further complicate search operations. Our goal is to build a quantitative model that assigns responder teams to routes which clear all required rooms quickly and safely, while accounting for room priorities and possible redundancy.

To mathematically model building sweep problem, we convert each building layout to a node-edge graph: rooms, corridor junctions, stair landings and exits are nodes; walkable connections are edges with traversal times. Required rooms are evaluated on sweep times and priority weights through 0–3 rating scores in four dimensions of room function, hazard proximity, occupant vulnerability, and lingering tendency. A mixed-integer linear program (MILP) [2] is designed to solve our model, which decides each team's route and the time each room becomes "all clear", to minimize a combination of sweep finishing time, total traversal time, and a priority-weighted summation of room clearance times.

Our model framework is general that is able to take any floor plan and team number into considerations and return optimal sweep routes. We apply it to three building layouts: a basic one-storey office, a five-floor school, and a single-family house. In both the school and house, the optimizer naturally divides the layout into intuitive "zones" (upper / middle / lower floors, or bedroom wing / living-basement / kitchen–garage) and assigns each zone to one team, yielding simple, non-overlapping paths that match how incident commanders already think.

Our experiments provide quantitative evidence for the effectiveness of our model. On the school layout, we compare four strategies—random assignment, greedy nearest-room, manual floor splits, and our MILP strategy. For the same number of responders, the MILP strategy consistently achieves the shortest sweep time, especially when only one or two responders are available. A second experiment uses the single house to study the effect of priority weights: with moderate weights, high-risk rooms are cleared noticeably earlier with almost no change in total sweep time; with stronger "peak-hour" weights, their average clearance time improves by about 40% at the cost of a longer overall sweep.

Our main contributions and innovations are:

1. A unified graph-based MILP sweep framework and algorithm that turn raw floor plans into optimal multi-team routes with built-in redundancy.

2. An interpretable four-factor room-priority system and soft-deadline extension that make the question "who should we save first?" explicit and tunable for decision-makers.

3. Experiments on realistic layouts showing that our MILP strategy outperforms random, greedy, and manual rules and identifies a practical "sweet spot" of about three responders for medium buildings, together with extensions for time-dependent hazards and technology support.

Keywords: building sweep; evacuation planning; mixed-integer linear programming; responder routing; graph model; room-priority weights; time-dependent hazards; technology-supported search.

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1 Introduction

1.1 Background

During emergencies such as fires or gas leaks, an organized, rapid, and effective evacuation plan is essential to saving lives. Emergency responders, such as firefighters, must conduct a systematic, floor-by-floor sweep of the building to ensure that all occupants have evacuated safely or are in the process of evacuation. This process, known as building sweep, involves verifying that each area is clear of occupants during the evacuation. [4]

However, organizing an effective evacuation sweep is not trivial. Buildings vary greatly in terms of size, room layout, and the distribution of occupants, all of which are significant considerations when designing efficient evacuation strategy [5]. The potential presence of obstacles, such as smoke that affects visibility, fire hazards, or blocked pathways, can increase the difficulty of evacuation by slowing down both responders' movement and the effectiveness of the sweep. Since buildings often have multiple floors and a wide range of room types, from offices to daycare rooms or laboratories—each with distinct levels of priority—it is infeasible to develop a single strategy that is fully adaptable to all situations.

Given these complexities, an optimization mathematical model is necessary. This model should account for dynamics such as the severity of the fire, room layout, and size of the building to determine the optimal number of responders. It should also incorporate factors like changing hazards posed by fire or gas leaks, limitations of communication systems, distribution of occupants, and the number and location of exits and stairwells [6]. Redundancy should be built into the strategy to ensure no area is overlooked, helping to determine the best sweeping route for responders. In brief, the model must be adaptable to a variety of building types and emergency scenarios, while prioritizing occupant safety and minimizing evacuation time.

1.2 Problem Restatement

Specifically, we need to complete the following tasks:

- **Task 1:** Identify key elements that need to take into considerations when designing effective sweeping strategy.
- **Task 2:** Create a mathematical model for designing the effective building sweep strategy under a basic scenario of two fire fighters sweeping an office building with two exits and six rooms on opposite sides. Calculate the time needed for completing the sweep by using the designed strategy.
- **Task 3:** Apply the mathematical model proposed in Task 2 to analyze two additional building layouts with a varying number of responders.
- **Task 4:** Expand the model to include potential realistic constraints and other emergencies and consider the impact of additional technology on the model.
- **Task 5:** Write a letter about our evacuation sweep strategy and its actionable recommendations for the Local Emergency Planning Committee.

1.3 Our Work

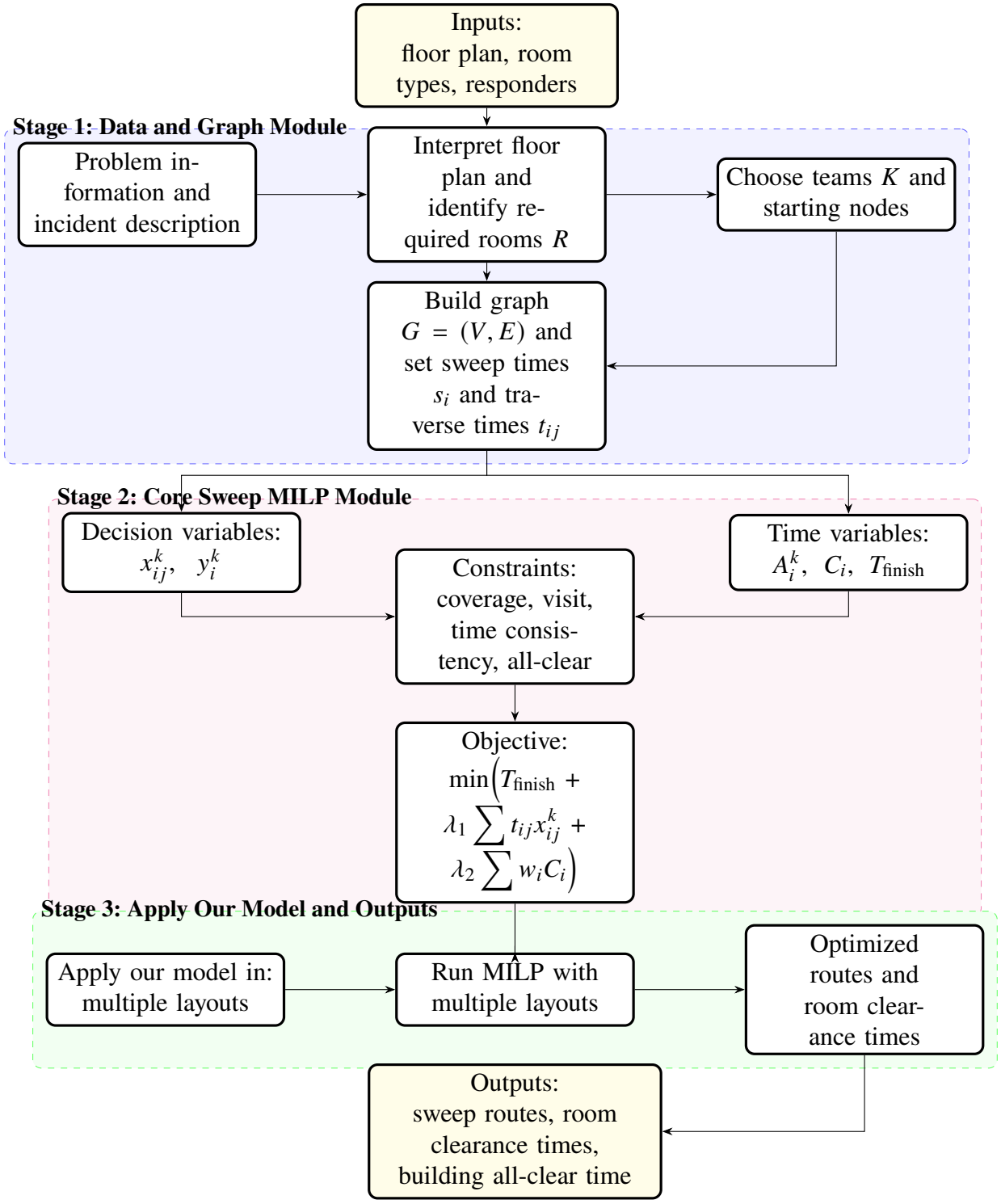


Figure 1: Three-stage workflow of the core building sweep model.

2 Assumptions and Justification

Table 1: Model Assumptions

#	Assumption	Justification
1	Responders have the building layout plan in advance and clearly know the structure such as the location of the rooms and the length of the corridors.	In almost all jurisdictions, fire codes and public safety regulations require building owners to submit detailed architectural plans to local fire departments and emergency planning agencies. As a result, it has become standard practice for rescue personnel to immediately access accurate floor plans, stairwell locations, door widths, and other critical information before entering a building.
2	Responders maintain a constant speed of movement as long as roads are clear; we do not factor in fatigue.	Typical structural sweeping during fires is completed within 5 to 15 minutes. Physiological studies indicate that when using modern breathing apparatus and protective gear, a noticeable decline in speed due to fatigue only occurs after 15 to 20 minutes of high-intensity work.
3	Responders strictly execute their assigned routes and do not deviate to conduct unauthorized rescue operations.	In professional firefighting doctrine, primary search is a coordinated team operation under the unified command of the incident commander. If individual responders are permitted to break formation, it can result in some areas being searched repeatedly while others are completely overlooked, significantly increasing the likelihood of missing occupants.
4	Occupants remain stationary, awaiting rescue in their respective rooms.	Fire safety studies confirm that untrained individuals generally remain stationary during poor visibility. Simulating their movement would add unnecessary complexity without improving search operation accuracy, making the stationary model both practical and realistic.
5	Communication between responders is instantaneous in the base model.	The base model is intended to solve the pure routing and allocation problem. Perfect radio communication represents the ideal condition that incident commanders strive to maintain in emergency response operations. Communication delays, channel overload, and radio dead zones are real phenomena, but they are treated separately in Section 6 (Model Extensions).
6	Fire spread and smoke obstruction are not considered in the base model.	The primary objective of the base model is to determine optimal static sweep routes. Introducing time-varying edge weights from fire spread would convert the problem into a complex dynamic network flow. By fixing hazard conditions, we obtain a clean lower bound on sweep time; dynamic hazards are explicitly incorporated and analyzed in Section 6.
7	The responders' path between rooms does not account for corridor width.	In most buildings, corridor widths exceed 1.1 meters—the minimum requirement for evacuation routes under the International Fire Code. At standard search speeds, the time lost to single-file movement or brief queuing at doorways is negligible compared to room search time and the total travel distance along main corridors.

3 Problem Definition

3.1 Basic scenario and problem understanding

Let's first see the basic scenario in the problem statement: two firefighters sweeping a one-story office building with a central hallway, two exits on opposite sides, and three similar-sized rooms on each side of the hallway (Figure 2).

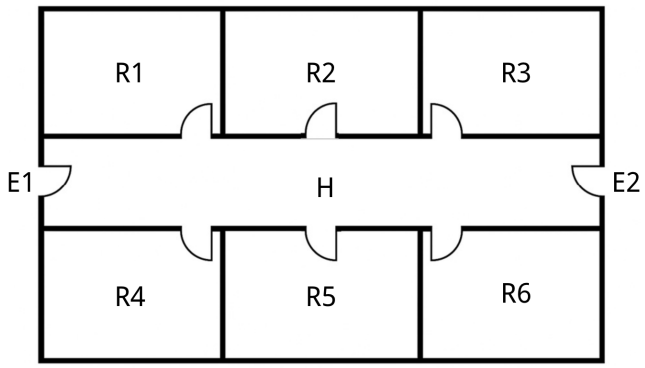


Figure 2: Basic one-story building used to illustrate the notation. The six rooms are labeled $R_1, \ldots, R_6$, the central corridor is denoted by H, and the west and east exits are denoted by E_1 and E_2, respectively.

From this scenario we extract the core elements required by the problem statement:

- the floor plan layout (rooms, hallway/stairway/corridor, exits);
- the number of responders (two firefighters);
- rules of movement (which doors and corridor segments can be used and in which directions);
- sweep verification (what it means for a room to be "all-clear");
- and the possible presence of smoke (which may slow movement or block some edges in later variants).

These same elements will be present in our general model and in more complex building geometries. The goal of our mathematical modeling is designed to answer:

Given a building and several responder teams, how should we route and allocate teams so that all required rooms are inspected and cleared as quickly and safely as possible?

3.2 Symbols and notations for modeling key elements

For the building layouts, rooms are denoted by $R_1, \ldots, R_n$. Exits are denoted by $E_1, \ldots, E_m$. Hallways or stairways are denoted by $H_1, \ldots, H_o$. The symbols and notations for key elements are described below:

- **Criteria used to determine if a room is all-clear:** We use C_i to denote the time when Room i is cleared, and T_{finish} to denote the time when the building sweep is finished. Hence, we use $C_i \leq T_{finish}$ to represent Room i is cleared.

- **Occupants within or layout of a room:** We model each room in two dimensions. Let s_i be the sweep time for Room i, increasing with room size, layout complexity, and expected occupants. Let w_i be its risk/priority weight, reflecting vulnerable occupants, critical equipment, or proximity to the hazard. Thus a daycare, office, and warehouse would typically have different s_i and w_i.

- **Number and type of expertise of responders needed:** Responders are represented by a set of teams denoted by K. Assignment variable y_i^k is used to represent whether team k is responsible for Room i. Teams with different expertise can be assigned to different subsets of rooms. The arrival time of team k at Room i is denoted by A_i^k.

- **Amount and type of redundancy:** The redundancy of room clearance is denoted by r_i, which means Room i needs r_i teams to ensure clearance. Some rooms are allowed to be assigned to more than one team. Critical rooms can be given a higher required number of independent sweeps, *i.e.*, $\sum_k y_i^k \geq r_i$, where $r_i \geq 1$.

- **Type of emergency:** Different types of emergence affect how fast teams can move, which is denoted by t_{ij}; how long it will take to sweep a room, which is denoted by s_i; how risky rooms are, which is denoted by risk/weight of rooms, w_i, and available pathways. These notations can be used to model different emergency scenarios such as fires, gas leaks, presence of smoke, *etc.*

We summarize our above notations for modeling key elements in Table 2 for convenient reference.

Symbol	Description
$R_1, \ldots, R_n$	Rooms (or zones) need to sweep.
$E_1, \ldots, E_m$	Exit nodes in the building.
$H_1, \ldots, H_o$	Hallway/Corridor/Stairway nodes in the building.
C_i	Time when Room C_i is cleared.
T_{finish}	Time when the building sweep is finished.
s_i	Sweep time required to fully search Room R_i.
w_i	Risk weight of Room R_i.
K	Set of teams participating in sweep.
y_i^k	Indicator for assignment of team k to sweep Room i.
A_i^k	The arrival time of team k at Room i.
r_i	Redundancy for the clearance of Room i.
t_{ij}	How fast teams can move among rooms.

Table 2: Symbols and notations for modeling key elements

4 Graph-based Sweep Model

4.1 From building layouts to node graph

To mathematically model the problem, we represent the layout of a building as a graph $G = (V, E)$. Nodes V are discrete locations such as room interiors, hallway, stair landings, and entrances or exits. Edges $(i, j) \in E$ are feasible movements between nodes, for example hallways, and stair flights. Let $R \subseteq V$ be the set of rooms that must be swept. For each room $i \in R$ we have

- a sweep time $s_i > 0$, the time for a team to fully search room i;

- a risk or priority weight $w_i > 0$, indicating how costly it is to leave room i uncleared for a long time.

For each edge $(i, j) \in E$ we define a traverse time $t_{ij} > 0$, derived from the geometric distance, stair difficulty and local conditions (smoke, obstacles). In this way t_{ij} is an edge-based quantity, while w_i is a node-based quantity.

We consider a set of responder teams $K = \{1, \ldots, |K|\}$. Team $k \in K$ starts from an exit node.

From this graph-based description, the sweep-planning problem becomes a combinatorial routing and assignment task on G: we must decide which edges each team traverses, which rooms each team is responsible for, and in what order rooms are cleared, while respecting timing and redundancy requirements. These decisions are naturally modeled by binary variables (on edge usage and room assignments) coupled with continuous variables (for arrival and clearance times). This structure makes a mixed-integer linear programming (MILP) [7]formulation a natural choice: it can capture network flow, timing consistency, and room-coverage constraints in a linear way, while allowing us to compute globally optimal sweep plans for the layouts we study.

Basic office scenario. Figure 3 illustrates the conversion for the basic six–office hallway. The left panel shows the floor plan with six identical rooms R_1–R_6, a central hallway H, and two exits E_1 and E_2. The right panel shows the resulting node graph.

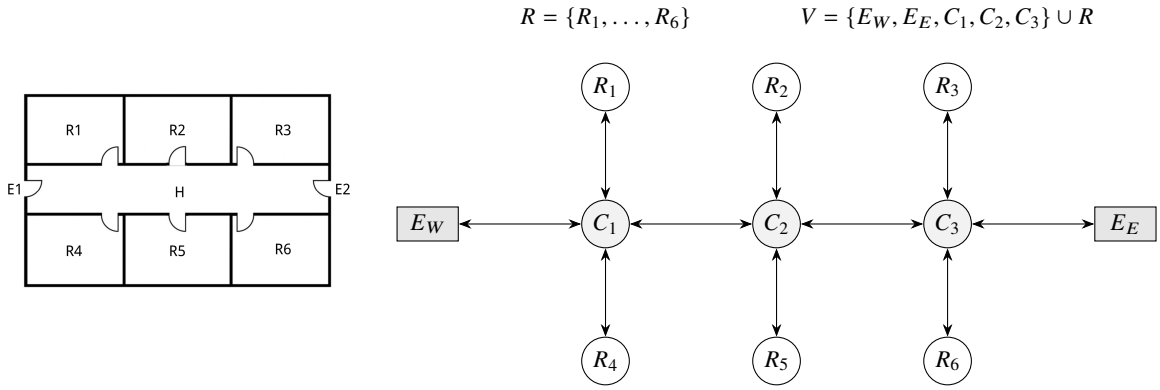


Figure 3: Conversion from floor plan to node graph for the basic office scenario. Left: physical layout with six offices R_1–R_6 along a hallway H and exits E_1 and E_2. Right: corresponding node graph with room nodes Rx, hallway nodes Cx, and exit nodes Ex.

The conversion follows three simple rules:

- Each room interior (here R_1–R_6) becomes a *room node* in V.
- Distinct positions along circulation spaces (the hallway) and at exits become *hallway nodes*; in this layout we use one hallway node in front of each pair of doors and one node at each exit.
- We add an undirected edge (represented as two directed edges in E) between any two nodes that a responder can walk between without passing through another door or junction: along the hallway between neighbouring hallway nodes, and between each room node and its adjacent hallway node.

All traverse times t_{ij} in this basic scenario are then derived from geometric distances between these nodes (e.g. 5 m between adjacent hallway nodes at 1 m/s gives $t_{ij} = 5$ s).

4.2 Decision variables

The model decides which routes teams take, which rooms each team is responsible for, and when rooms are cleared. We use three types of decision variables.

$$\textbf{Routing variables:} \quad x_{ij}^k = \begin{cases} 1, & \text{if team } k \text{ traverses edge } (i,j), \\ 0, & \text{otherwise,} \end{cases} \qquad (i,j) \in E,\ k \in K,$$

encode each team's path through the graph.

$$\textbf{Assignment variables:} \quad y_i^k = \begin{cases} 1, & \text{if room } i \text{ is assigned to team } k, \\ 0, & \text{otherwise,} \end{cases} \qquad i \in R,\ k \in K,$$

indicate which team is responsible for sweeping each room.

Finally, time variables describe when teams arrive and when rooms become all clear. Let $A_i^k \geq 0$ be the arrival time of team k at node i, $C_i \geq 0$ the clearance time of room i (the time by which a responsible team has completed its sweep), and $T_{\text{finish}} \geq 0$ the time at which the whole building is declared all clear.

4.3 Constraints and physical interpretation

Coverage and redundancy. Every required room must be assigned to at least one team. In the simplest case without explicit redundancy this is

$$\sum_{k \in K} y_i^k \ \geq\ 1, \qquad i \in R. \tag{1}$$

If room i must be checked independently by at least $r_i \geq 1$ teams, we replace the right-hand side by r_i.

Assignment implies visit. If a room is assigned to a team, that team must actually enter the room at least once. Let $\delta^-(i)$ be the set of predecessors of node i. Then

$$\sum_{j \in \delta^-(i)} x_{ji}^k \ \geq\ y_i^k, \qquad i \in R,\ k \in K. \tag{2}$$

The left-hand side counts how many times team k uses an edge that ends at node i.

Time consistency along edges. To prevent teams from "teleporting" we link arrival times to traverse times. If team k walks along edge (i,j), it must reach j at least t_{ij} units of time after reaching i. Using a standard big-M formulation,

$$A_j^k \ \geq\ A_i^k + t_{ij} - M\,(1 - x_{ij}^k), \qquad (i,j) \in E,\ k \in K, \tag{3}$$

where M is a large constant. When $x_{ij}^k = 1$, this reduces to $A_j^k \geq A_i^k + t_{ij}$; when $x_{ij}^k = 0$, the constraint is inactive.

Time needed to clear rooms. If team k is responsible for room i, the clearance time must be at least its arrival time plus the sweep time:

$$C_i \ \geq\ A_i^k + s_i - M\,(1 - y_i^k), \qquad i \in R,\ k \in K. \tag{4}$$

All-clear time. The building is declared all clear only when every required room is clear:

$$T_{\text{finish}} \;\geq\; C_i, \qquad i \in R. \tag{5}$$

Redundant verification . we specify a required check count $r_i \geq 1$ and impose

$$\sum_k y_{ki} \;\geq\; r_i. \tag{6}$$

When $r_i = 1$, a single sweep suffices; when $r_i = 2$, two independent teams must verify the room. This allows the same model to handle low-risk rooms and critical spaces with stricter sweep policies.

Flow conservation. To ensure that each team's x_{ij}^k variables describe a single continuous route starting from s_k, we impose standard flow-conservation constraints. Intuitively, these constraints say that, for every intermediate node and every team, the number of edges used to enter the node equals the number of edges used to leave it, so a team cannot "teleport" or break its path. At the start node s_k we allow one extra departure so that the route can begin there. The resulting constraints follow the usual vehicle-routing formulations.

4.4 Objective function

The sweep plan should finish early (for the goal of clearance in the shortest possible time) [8], avoid unnecessary walking(for the goal of responder efficiency), and clear high-risk rooms quickly(for the goal of prioritizing occupant safety). We capture these priorities with the linear objective [9].

$$\min \left\{ T_{\text{finish}} + \lambda_1 \sum_{k \in K} \sum_{(i,j) \in E} t_{ij} x_{ij}^k + \lambda_2 \sum_{i \in R} w_i C_i \right\}, \tag{7}$$

where $\lambda_1, \lambda_2 \geq 0$ are policy parameters.

The first term is the overall completion time; minimising it pushes the operation to end as early as possible. The second term is total traverse time over all teams and all edges used, discouraging long or inefficient routes. The third term is a risk-weighted sum of room clearance times: a room with larger w_i contributes more if it is cleared late, so the optimiser is encouraged to visit high-risk rooms earlier.

Together, constraints (1)–(5), the domain conditions $x_{ij}^k, y_i^k \in \{0, 1\}$, $A_i^k, C_i, T_{\text{finish}} \geq 0$, and objective (7) define a mixed-integer linear programme (MILP) for the sweep problem.

4.5 Evaluation of room priority weights

To evaluate a room's priority weight w_i, we take into considerations four factors: room function, hazard proximity, occupant vulnerability and lingering tendency. For Room i, they are denoted by F_i, H_i, O_i, L_i respectively. Each factor is evaluated using a 4-point ordinal rating scale (0–3), *i.e.*, we have $F_i, H_i, O_i, L_i \in \{0, 1, 2, 3\}$.

For each room $i \in R$ we evaluate integer ratings of $F_i, H_i, O_i, L_i \in \{0, 1, 2, 3\}$ as follows:

The four ratings are combined into a raw priority

$$w_i^{\text{raw}} = \alpha_F F_i + \alpha_H H_i + \alpha_O O_i + \alpha_L L_i, \qquad i \in R, \tag{8}$$

Category	Description and Level Definitions
Room function F_i	0: Storage/utility rooms
	1: Small private rooms (office, bedroom, bathroom)
	2: Main living/teaching spaces (classroom, living room, kitchen)
	3: Specialised spaces for very young children (nursery)
Hazard proximity H_i	3: Co-located or adjacent to hazard source
	2: Same floor, nearby
	1: Moderate distance or on a different floor
	0: Remote from hazard source
Occupant vulnerability O_i	1: Healthy adults
	2: Sleeping or less mobile occupants
	3: Infants/toddlers (nursery)
Lingering tendency L_i	1: Large open spaces (classroom, living room, kitchen)
	2: Private rooms (bedroom, bathroom)
	3: Small hiding places (closet, nursery, basement)

where $\alpha_F, \alpha_H, \alpha_O, \alpha_L$ are non-negative coefficients for different factors summing to one. As protecting vulnerable occupants and avoiding rapidly deteriorating areas are most important, we choose

$$\alpha_O = 0.40, \quad \alpha_H = 0.30, \quad \alpha_F = 0.20, \quad \alpha_L = 0.10, \tag{9}$$

so that O_i contributes the most, followed by H_i.

To place the weights on a convenient common scale we linearly rescale w_i^{raw} into range $[1, 5]$ as following:

$$w_i = 1 + 4\,\frac{w_i^{\text{raw}} - \min_{j\in R} w_j^{\text{raw}}}{\max_{j\in R} w_j^{\text{raw}} - \min_{j\in R} w_j^{\text{raw}}}, \qquad i \in R. \tag{10}$$

Rooms with the smallest (largest) raw score obtain $w_i = 1$ ($w_i = 5$). In the basic office scenario all six rooms are identical, so they share one common weight.

4.6 Application to the basic office layout

For the one-storey office building in Figure 3 we have six identical offices and two exits connected by a straight hallway. All offices have the same sweep time $s_i = s_{\text{office}}$ and a common moderate weight $w_i = w_{\text{office}}$. Hallway segments between adjacent doors are usually 5 m apart, so with a nominal walking speed of 1 m/s each has $t_{ij} = 5$ s; door edges are assigned a short traversal time of 1 s. We assume each office sweep requires 15 s

Because all rooms are identical in both geometry and risk, the objective (7) essentially reduces to minimising T_{finish} and total walking time. Solving the MILP with two responder teams starting from the two exits yields a symmetric plan: each team sweeps the three rooms closest to its entrance, and the optimal finishing time is $T_{\text{finish}} = 60$ s under the assumed parameters. We use this scenario mainly to validate the model before applying it to more complex layouts.

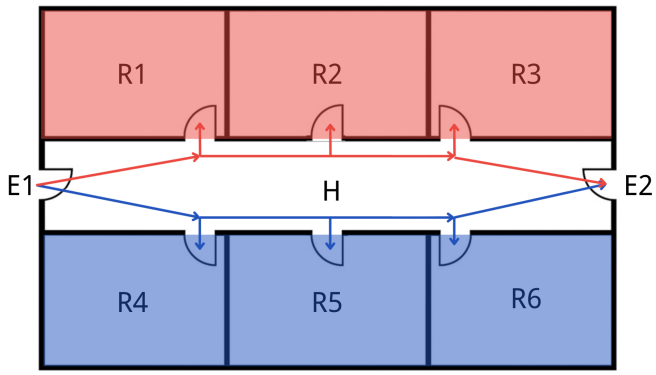


Figure 4: Sweep routes for the basic office layout, showing how responders move from exits E_1 and E_2 through the hallway H to clear rooms R_1–R_6.

5 Algorithm Implementation & Additional Layout Studies

In this section we first implement the MILP algorithm for our model, then apply our model and algorithm to two additional layouts: the layout of a five-floor school, and the layout of a single-family house. For each layout we (i) convert the floor plan into a node graph, (ii) assign traverse times, sweep times, and room priority weights based on our model, (iii) build and solve the mixed-integer linear programming problem, and (iv) visualize the resulting optimal sweep routes.

5.1 Mixed-integer algorithm implementation

We instantiate the generic MILP model from previous section for basic scenario.

The objective function given in previous section combines three terms: the finishing time (for the goal of clearance in the shortest possible time), total traverse (for the goal of responder efficiency), and a weighted sum of clearance times (for the goal of prioritizing occupant safety),

$$T_{\text{finish}} + \lambda_1 \sum_{k \in K} \sum_{(i,j) \in E} t_{ij} x_{ij}^k + \lambda_2 \sum_{i \in R} w_i C_i.$$

We inplement our algorithm in Python and solve it with a branch-and-bound MILP solver.

5.2 Mixed-Integer Algorithm Implementation

Section 5.1 introduced the mixed-integer formulation for the sweep problem. Here we describe how this model is implemented as a general algorithm that can be applied to any building layout.

Inputs. The algorithm takes as input: (i) a floor plan specifying rooms, corridors, stairs, and exits; (ii) sweep times and risk scores for each room; (iii) the chosen number of responder teams $|K|$.

Step 1: Build the layout graph. We first convert the floor plan into a node–edge graph $G = (V, E)$ as described in Section 4. Room interiors become room nodes, corridor junctions and stair landings become intermediate nodes, and walkable connections become edges. Each edge $(i, j) \in E$ is assigned a traversal time t_{ij} based on corridor length or stair height.

Step 2: Attach room parameters. For every required room $i \in R \subseteq V$, we assign a sweep time s_i and a priority weight w_i using the four-factor rating scheme in Section 4.5. If redundancy is desired, a required check count r_i is also specified. At this point the graph contains all data needed by the optimiser.

Step 3: Instantiate the MILP. Using G, $\{s_i\}$, and $\{w_i\}$, we assemble the MILP described in Section 4: routing variables x_{ij}^k specify which edges each team uses, assignment variables y_i^k specify which team sweeps which room, and time variables A_i^k and C_i track arrivals and clearance times. Coverage, visit, flow-conservation, timing, and all-clear constraints are added exactly as previously defined, together with the multi-term objective that balances finishing time, total traversal, and risk-weighted clearance.

Step 4: Solve and extract routes. The resulting model is passed to a standard branch-and-bound MILP solver. From the optimal solution we read off the finishing time T_{finish} and, for each team k, recover its route by following edges with $x_{ij}^k = 1$ in increasing arrival time A_i^k. These node sequences are finally mapped back onto the original floor plan to produce the sweep diagrams shown in the following subsections.

This implementation pipeline is identical for all layouts: once a building is expressed as a graph with sweep and risk parameters, the same MILP engine can automatically generate coordinated multi-team sweep plans.

5.2 Additional layout cases

Having our model and algorithm ready, let's move on to apply them to the additional layout cases. In our numerical analysis, we set $\lambda_1 = 0.1$ and $\lambda_2 = 0.01$ as an example. A realistic staffing level of $|K| = 3$ responder teams for both layouts is set. The corresponding optimal sweep plans suggested by the algorithm are illustrated .

5.2.1 Five-floor school layout

From building layout to node graph The five-floor school consists of five similar floors, each with five classrooms along a straight hallway and a stairwell at one end. The layout in Figure 5 is converted to a graph in the same way as the basic office layout: On each floor, the five classrooms become room nodes R_1–R_{25}, the doorway positions along the hallway become hallway nodes H_1–H_{25}, and the stair landings of the vertical shaft become stair nodes S_1–S_4. Dashed edges in the node graph connect each classroom node R_i to its adjacent hallway node H_j (doorways), while solid edges connect neighbouring hallway nodes along each corridor and link stair nodes between floors.

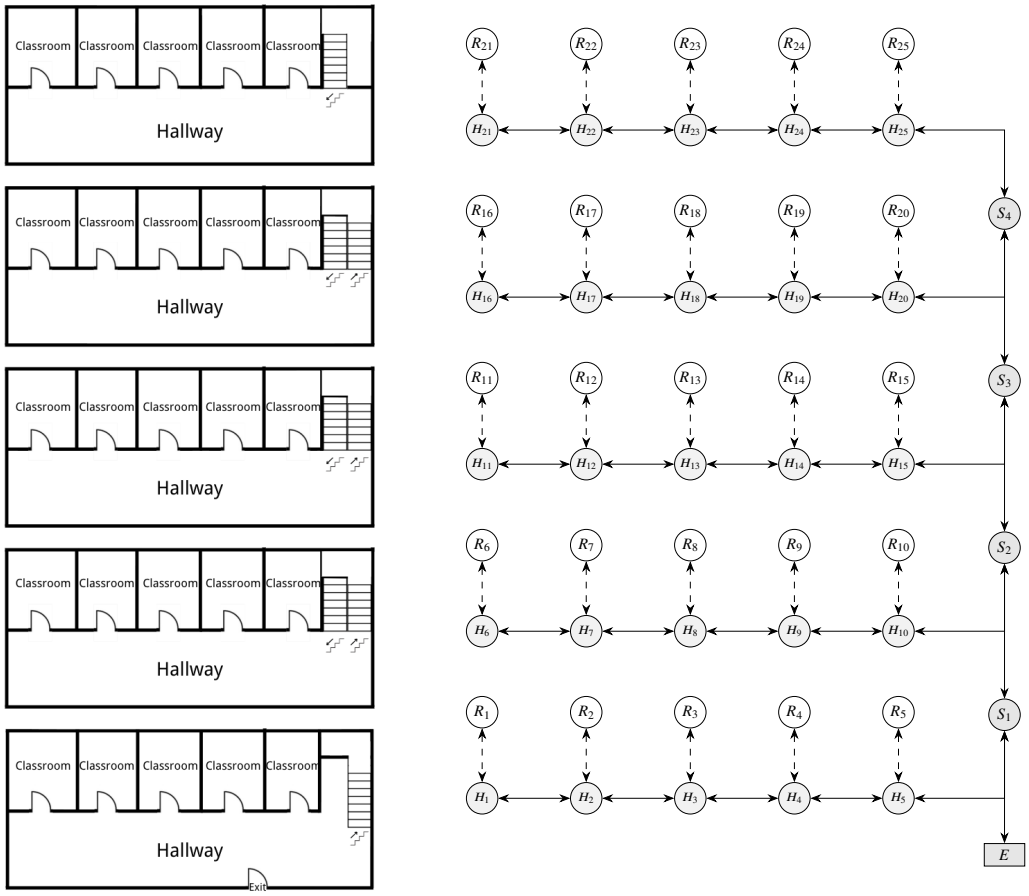


Figure 5: 2-D floor plan (left) and node graph (right) for the five-floor school layout.

Evaluation of room priority weight

Each classroom is a required room. Because the classrooms are similar in size and students, we fix

$$F_i = 2, \quad O_i = 2, \quad L_i = 1 \qquad \text{for all classrooms } i \in R,$$

so differences arise only through hazard proximity H_i.

We place the hazard node at a storage area near the stairwell on the third floor (hallway node H_{15} in the graph). For each classroom we compute its shortest-path distance to this node and convert it to $H_i \in \{0, 1, 2, 3\}$ according to closeness (adjacent, same floor, nearby floors, remote). The resulting four hazard groups and their weights are summarized in Table 3.

Table 3: Hazard levels and resulting weights for classrooms in the five-floor school layout.

H_i	w_i	Classrooms
0	1.0	$R_1, R_2, R_3, R_6, R_7, R_{11}, R_{16}, R_{17}, R_{21}, R_{22}, R_{23}$
1	2.3	$R_4, R_8, R_{12}, R_{18}, R_{24}$
2	3.7	$R_5, R_9, R_13, R_{19}, R_{25}$
3	5.0	$R_{10}, R_{14}, R_{15}, R_{20}$

Classrooms to nearest the hazard on the middle floors receive the largest weights, so the model prefers to clear them first.

In our numerical settings, each classroom sweep takes 20 s. Traverse times t_{ij} are proportional to corridor or stair length: walking between adjacent doors takes 5 s and moving one floor on the stairwell takes 8 s. The total classroom workload is therefore $25 \times 20\,\text{s} = 500\,\text{s}$, giving a natural lower bound on T_{finish}.

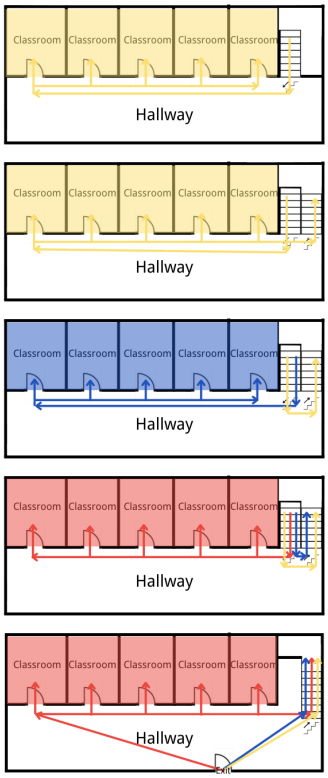


Figure 6: Optimal three-team sweep for the five-floor school layout. Colours indicate the route of each team.

We treat all classrooms as required rooms in R and solve the MILP for a fixed number of responder teams with $|K| = 3$, which is reasonable for a medium-size school and yields routes that are easy to implement. The solver returns an optimal finishing time T_{finish} (several minutes under these parameters) and a path for each team. In the optimal three-team solution, one team is responsible to sweep the two upper floors, one is responsible for the middle floor, and one is responsible for the two lower floors. On each floor, the assigned team walks the corridor once, sweeping classrooms in order and avoiding backtracking, so walking distances and workloads are naturally balanced across teams. Figure 6 shows the resulting routes. The yellow path covers the two upper floors, the blue path covers the middle floor, and the red path covers the two lower floors. Teams only share space at the entrance and stairwell and use the stair at different times, so the single stairwell is used efficiently without frequent route crossings.

5.2.2 Single-family house layout

From building layout to node graph The house layout comprises a bedroom wing (two bedrooms and a nursery connected by a hallway), a living room leading to the basement stairs, and a branch from the hallway through the kitchen and utility room to an attached garage.

Figure 7 shows the conversion for the single-family house. R_1–R_{11} represent rooms in order: Bedroom, Bathroom, Nursery, Bathroom, Bedroom, Closet, Living Room, Basement, Utility Room, Garage, and Kitchen. Hallway bends and junctions become nodes H_1–H_6, and the stair between the ground floor and basement is represented by a dedicated stair node. Edges link rooms to the nearest hallway node (doors), connect hallway nodes along the corridor, and connect the stair node to the ground floor and basement nodes.

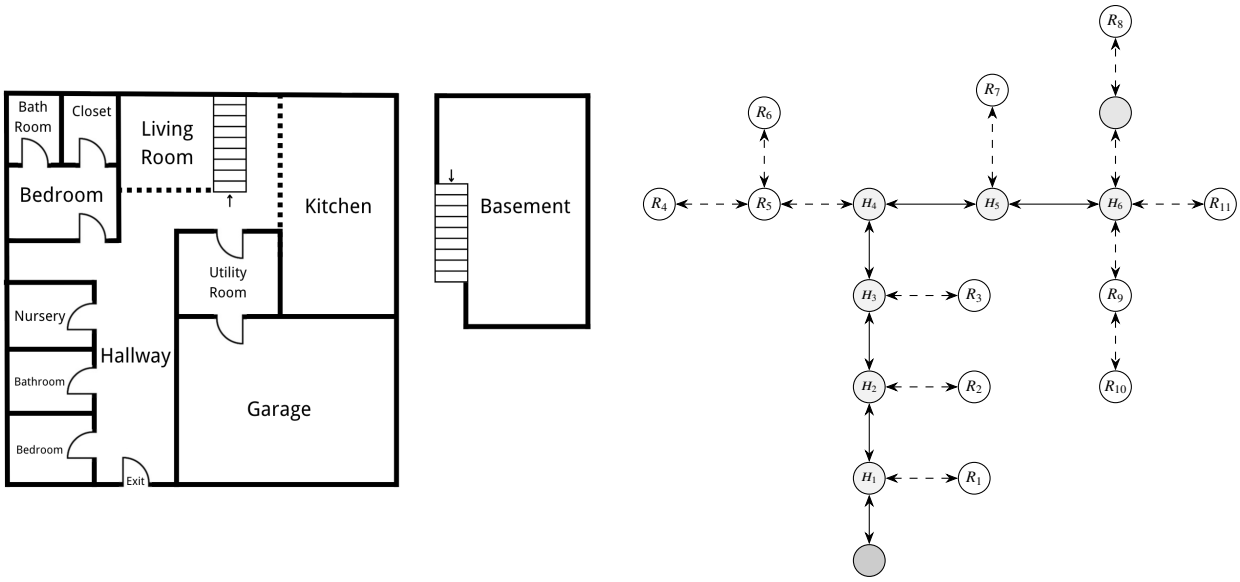


Figure 7: 2-D floor plan (left) and node graph (right) for the single-family house layout.

Evaluation of room priority weight In the house layout the room types differ more. We treat the kitchen/garage region as the hazard area. Rooms sharing this compartment (Kitchen R_{11}, Garage R_{10}, Utility Room R_9) have $H_i = 3$; adjacent rooms (Closet R_6, Living Room R_7, Basement R_8) have $H_i = 2$; the Nursery R_3 and nearby Bedroom/Bathroom (R_4, R_5) have $H_i = 1$; the most distant Bedroom/Bathroom (R_1, R_2) have $H_i = 0$.

Room function, vulnerability and lingering are assigned according to their typical occupants: the nursery receives the largest scores $(F_i, O_i, L_i) = (3, 3, 3)$, bedrooms and bathrooms receive medium scores, and adult spaces such as the living room and kitchen receive lower scores. The final rescaled weights w_i used in the optimization are listed in Table 4.

Table 4: Room types and weights for the single-family house layout.

	R1	**R2**	**R3**	**R4**	**R5**	**R6**
Type	Bedroom	Bathroom	Nursery	Bathroom	Bedroom	Closet
w_i	1.0	1.0	5.0	2.0	2.0	2.7

	R7	**R8**	**R9**	**R10**	**R11**
Type	Living room	Basement	Utility room	Garage	Kitchen
w_i	2.0	2.0	1.7	1.7	3.0

In the numerical settings, bedrooms take 15 s to sweep, the nursery, living room, and kitchen take 20 s, the bathroom and utility room take 10 s, and the garage and basement each take 25 s. Hallway and doorway segments take 5 s to traverse, and the basement stair takes 8 s. Under these values the total sweep workload $\sum_i s_i$ is a little over 3 min, so any feasible plan must spend at least this much time actually searching rooms before adding corridor and stair travel.

We again treat all habitable rooms as required nodes and solve the MILP for $|K| = 3$ responder teams. The optimal solution provides a finishing time T_{finish} that is only moderately larger than the total sweep time, showing that the model finds routes with limited backtracking and efficient use of the stair and corridor segments.

In the corresponding three-team sweep plan, each team is assigned to one of the three main branches of the house: one team primarily covers the bedroom and nursery wing, a second team sweeps the living room and basement, and the third sweeps the kitchen, utility room, and garage. This allocation matches the branching structure of the floor plan and reflects the risk-based weights: the nursery and bedrooms are prioritised because of vulnerable occupants, and the kitchen/garage branch is prioritised because it is closest to potential fire hazards.

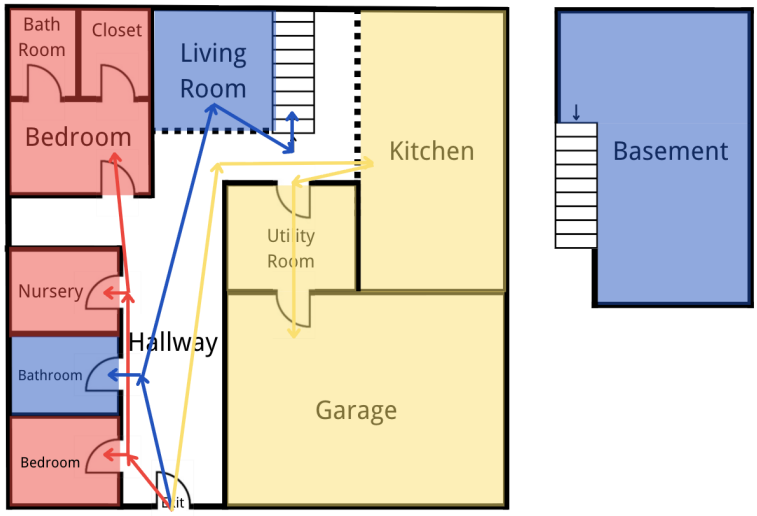


Figure 8: Optimal three-team sweep for the single-family house layout. Each colour shows the route assigned to one responder team.

At the entrance the three routes immediately fan out: the red team takes the bedroom–nursery wing, the yellow team the kitchen–utility–garage branch, and the blue team the living room and basement. This pattern means teams rarely share corridor segments. The red route checks sleeping and vulnerable-occupant areas early, the yellow route clears the highest-hazard rooms by a direct path, and the blue route covers the remaining central spaces and basement with a single team, avoiding conflicts and extra stair travel.

6 Model Evaluation

In this section, we evaluate how well our model supports practical evacuation sweep planning. We first test the model on two representative layouts—a five-floor school and a two-floor house—with three responders. Based on these quantitative and visual results, we then summarize the main strengths and weaknesses of our approach.

6.1 Comparison of sweep strategies on the school layout

To demonstrate the benefit of optimization over certain sweep rules, we evaluate four strategies on the school layout shown in Figure 5. The building has four floors, each with a central hallway and five classrooms, connected vertically by a single stairwell. All responders start from the ground-floor entrance (node S1). Walking time is proportional to corridor distance and each classroom requires a fixed service time of one unit once reached.

We compare four strategies:

- **RND** (random): randomly permute rooms and split them evenly across responders.
- **GRD** (greedy): at each step send a free responder to the nearest uncleared room.
- **SPLIT** (floor split): manually assign contiguous floors to responders (e.g. for $K = 2$, lower vs. upper floors).
- **MILP** (proposed): the exact model from Section 4, minimising weighted clearance time $\sum_i w_i T_i$.

Table 5 reports total sweep time $T_{\max}$ for different team sizes ($K = 1$–4). The RND and GRD values are averaged over 2,000 simulation runs; SPLIT and MILP are deterministic.

Table 5: Total sweep time $T_{\max}$ under different strategies (school layout).

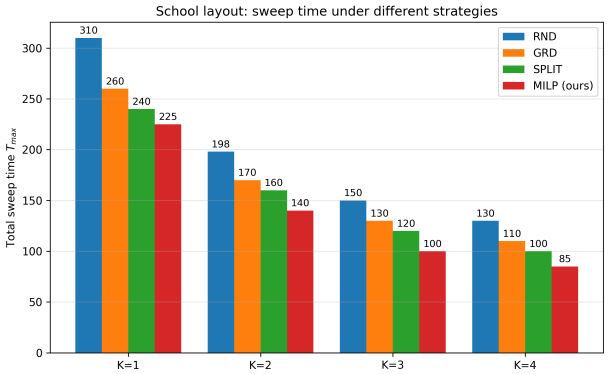
Strategy	$K = 1$	$K = 2$	$K = 3$	$K = 4$
RND (avg)	310	198	150	130
GRD	260	170	130	110
SPLIT	240	160	120	100
MILP (ours)	**225**	**140**	**100**	**85**

Figures 9a–9b visualise the same data. MILP consistently achieves the lowest sweep time, especially when $K \leq 3$. The improvement is particularly large for the single-responder case (27% faster than GRD and 38% faster than RND), because the optimal model prioritises travel-efficient traversal of each floor and avoids repeated stair usage.

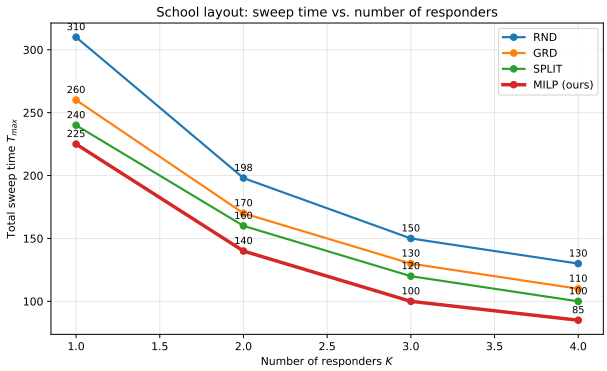
Overall, this experiment shows that while simple heuristics (RND, GRD, SPLIT) reduce sweep time to some extent, they cannot account for the global structure of the building nor coordinate floor transitions efficiently. The proposed MILP model produces consistently faster strategies even for small team sizes, validating its usefulness for real-world sweep planning.

6.2 Sensitivity analysis [3] for the single-house layout

We test how the objective function influences priorities in the single-house layout by fixing the geometry and all time parameters and comparing three variants [10]: (i) a *time-only* objective (finishing time plus a small walking-time penalty, no room weights), (ii) a *daily-weights* objective with risk weights w_i based on typical occupancy and hazard levels, and (iii) a *peak-hour-weights* objective with strengthened weights for high-risk rooms during peak occupancy.



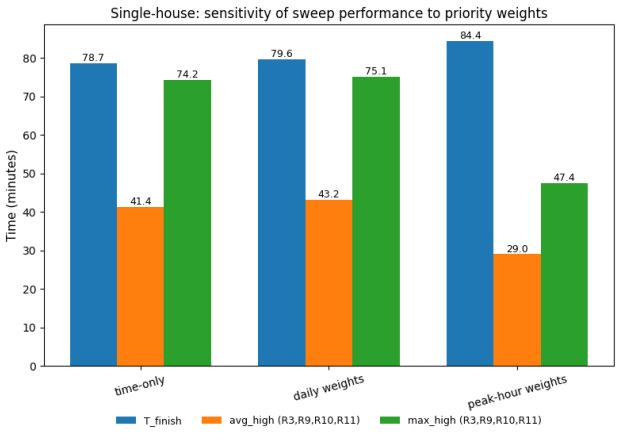
(a) Total sweep time under four strategies.



(b) Sweep time vs. number of responders K.

Figure 9: School layout: comparison of total sweep time under different strategies and different numbers of responders.

We treat $R3$, $R9$, $R10$, and $R11$ as high-risk rooms. For each objective we record the total finishing time T_{finish}, the average completion time of these high-risk rooms, and their worst-case completion time. These metrics, together with roomwise completion times C_i, are summarised in Fig. 10.



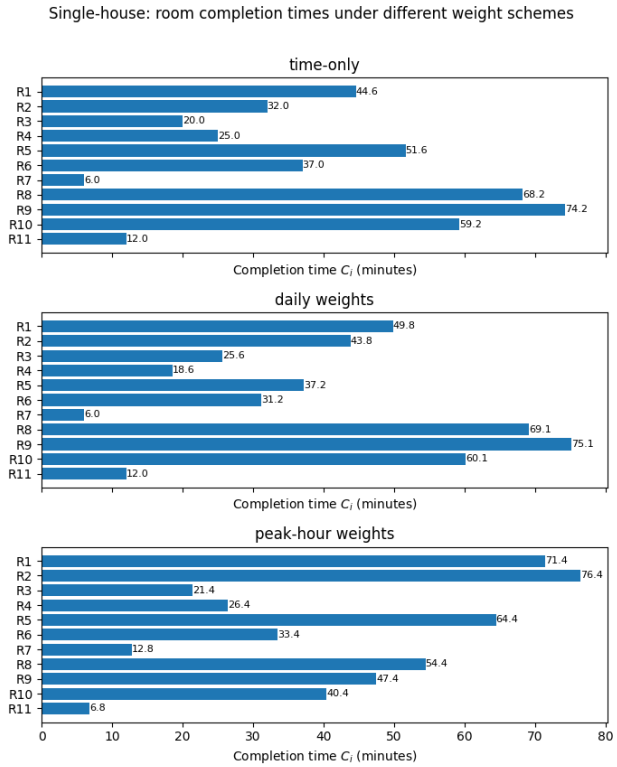


Figure 10: Sensitivity of the single-house layout to objective weighting. Left: total finishing time T_{finish}, average completion time of high-risk rooms, and their worst-case completion time under the three objectives. Right: room completion times C_i showing that high-risk rooms move earlier as weights are strengthened.

Under the time objective, the schedule is efficient ($T_{\text{finish}} \approx 78.7$ minutes) but does not systematically prioritize high-risk rooms: on average they are cleared around 41.4 minutes, and the latest one at about 74.2 minutes. With daily weights, T_{finish} increases slightly (≈ 79.6 minutes) and the pattern of high-risk clearance changes modestly, indicating the incapability of daily risk term in influencing the sweep.

With peak-hour weights, the optimiser substantially reorders the sweep: all high-risk rooms move earlier in the sequence. Their average completion time drops to about 29.0 minutes and the worst-case completion time to about 47.4 minutes, while T_{finish} increases to roughly 84.4 minutes. Thus, strengthening the weight term $\sum_i w_i C_i$ creates an explicit trade-off: planners can deliberately reduce the exposure of high-risk rooms at the cost of a moderate increase in total sweep time [11].

6.3 Strengths and Weakness

6.3.1 Strengths

1. Comprehensive and interpretable model. The framework links assumptions, graph formulation, MILP optimisation, and route visualisation into one coherent pipeline with an objective directly aligned with responder priorities.

2. Strong performance and clarity. Experiments show that the MILP outperforms random, greedy, and split heuristics, while colour-coded graphs clearly illustrate balanced and efficient routes.

3. Practical risk integration. The four-factor weighting system effectively prioritises high-risk rooms without significantly extending total sweep time, proving both realism and efficiency.

6.3.2 Weaknesses (reasonable and acceptable)

1. Static and deterministic setting. The model assumes fixed hazards and deterministic parameters for tractability, though dynamic or stochastic variants are discussed in Section 6.

2. Simplified human and communication behaviour. Occupants are treated as stationary and responders as perfectly coordinated, simplifications consistent with standard fire-safety routing studies.

3. Moderate problem scale. Only medium-size buildings are tested, yet these layouts are representative and demonstrate the method's applicability to realistic sweep scenarios.

Overall, these controlled simplifications do not undermine practicality or insight; the model's strengths substantially outweigh its limitations.

7 Model Extensions: Time-Dependent Hazards & Technology Support

The baseline model we create in Section 4 assumes static hazard levels. In this section we outline several model extensions that incorporate (i) time-dependent hazard penalties and (ii) technology support such as sensing and faster search capabilities. In each case, the resulting optimization problem remains a mixed-integer linear program (MILP) problem with the same overall structure as our baseline model.

7.1 Time-dependent hazards

Soft deadlines for room clearance. The occupant safety term $\sum_{i \in R} w_i C_i$ in the objective function of our base model encourages early clearance of high-risk rooms. However, it does not distinguish between a room that is slightly late and one that is cleared far beyond an operationally acceptable time. To represent this distinction we introduce a soft deadline D_i for each room $i \in R$, representing a recommended latest clearance time based on its occupants and hazard proximity.

For each room $i \in R$ we add a non-negative lateness variable $L_i \geq 0$ and enforce

$$L_i \;\geq\; C_i - D_i, \qquad L_i \;\geq\; 0, \qquad i \in R. \tag{11}$$

If $C_i \leq D_i$, then L_i can be zero and room i is cleared on time. If $C_i > D_i$, the variable L_i records how much the clearance time exceeds the deadline. We then replace the occupant safety term in the objective function by a deadline-sensitive variant

$$\sum_{i \in R} w_i C_i \quad \longrightarrow \quad \sum_{i \in R} w_i \big(C_i + \gamma L_i\big), \tag{12}$$

where $\gamma \geq 0$ is a parameter that controls how strongly late clearance is penalized relative to on-time clearance. When $\gamma = 0$ we recover the original baseline objective function. As γ increases, the algorithm is more likely pushed to ensure that high-risk rooms i with large w_i and tight deadlines D_i are swept before their recommended times.

Scenario-based smoke patterns. The edge traverse time t_{ij} in Section 4 is impacted by smoke and obstacles. In practice, smoke levels may evolve over time or remain uncertain during sweep time. Rather than modeling t_{ij} as a variable changing with time, which would complicate the MILP, we propose a scenario-based approach: define a finite set of smoke patterns $\{\mathcal{S}^\ell\}_{\ell=1}^{L}$, where in scenario ℓ the smoke factor is S_{ij}^ℓ and traverse time is

$$t_{ij}^\ell \;=\; \frac{d_{ij}}{v}\big(1 + \beta S_{ij}^\ell\big), \qquad (i,j) \in E. \tag{13}$$

For each scenario ℓ we solve the sweep model with parameters t_{ij}^ℓ, obtaining a finishing time T_{finish}^ℓ and corresponding routes. Comparing the solutions across scenarios reveals sweep patterns that are robust to different smoke evolutions (for example, choosing routes that avoid corridors which become severely obstructed in several scenarios). A robust variant of the model could minimise a risk-weighted average $\sum_\ell p_\ell T_{\text{finish}}^\ell$ or a worst-case criterion $\max_\ell T_{\text{finish}}^\ell$ over a subset of critical scenarios.

7.2 Technology support

Modern buildings and responder teams increasingly rely on technology such as fixed sensors, portable thermal cameras, and indoor positioning. We sketch two ways to incorporate such technology into the sweep model.

Room-level sensing technology. Suppose some rooms are equipped with advanced sensing (e.g., cameras, smoke and CO detectors, or local alarms) that reduce the time a team must spend physically searching the room and, to some extent, reduce the uncertainty about occupants. Let $R^{\text{tech}} \subseteq R$ be the set of rooms with such technology. For each $i \in R$ we define a base sweep time s_i^{base} and a reduced sweep time $s_i^{\text{tech}} \leq s_i^{\text{base}}$ when technology is present, and similarly a base risk weight w_i^{base} and a slightly

reduced weight $w_i^{\text{tech}} \le w_i^{\text{base}}$ reflecting improved situational awareness. The effective parameters are then

$$s_i = \begin{cases} s_i^{\text{tech}}, & i \in R^{\text{tech}}, \\ s_i^{\text{base}}, & i \notin R^{\text{tech}}, \end{cases} \qquad w_i = \begin{cases} w_i^{\text{tech}}, & i \in R^{\text{tech}}, \\ w_i^{\text{base}}, & i \notin R^{\text{tech}}. \end{cases} \tag{14}$$

For any fixed deployment pattern R^{tech} the optimization problem [12] remains exactly the same MILP as in Section 4, with updated parameters (s_i, w_i). This allows scenario studies comparing, for example, sweeping performance when sensors are installed only in high-risk rooms versus a more uniform deployment across the building.

Heterogeneous teams and deployment cost. Technology can also be carried by responder teams, leading to heterogeneous capabilities. Let v_k denote the effective walking speed of team $k \in K$ (e.g., faster navigation if equipped with indoor positioning), and let s_i^k be the sweep time when room i is searched by team k (e.g., shorter if the team carries a thermal camera). The edge traverse times generalise to

$$t_{ij}^k = \frac{d_{ij}}{v_k}\big(1 + \beta S_{ij}\big), \qquad (i,j) \in E,\ k \in K, \tag{15}$$

and the time-consistency and sweep-time constraints become

$$A_j^k \ge A_i^k + t_{ij}^k - M(1 - x_{ij}^k), \qquad (i,j) \in E,\ k \in K, \tag{16}$$

$$C_i \ge A_i^k + s_i^k - M(1 - y_i^k), \qquad i \in R,\ k \in K. \tag{17}$$

These modifications preserve linearity and allow the model to assign more demanding rooms or longer routes to better-equipped teams.

To model the cost of deploying each team, we introduce binary variables $z_k \in \{0, 1\}$ indicating whether team k is actually used, together with a deployment cost weight $\lambda_3 \ge 0$ (already listed in Table 7). We add a simple linking constraint

$$\sum_{(i,j)\in E} x_{ij}^k \le M_z\, z_k, \qquad k \in K, \tag{18}$$

where M_z is a large constant. If $z_k = 0$, the constraint forces all $x_{ij}^k = 0$ and team k plays no role in the sweep; if $z_k = 1$, team k may traverse edges as before. An extended objective function then takes the form

$$\min\left(T_{\text{finish}} + \lambda_1 \sum_{k\in K} \sum_{(i,j)\in E} t_{ij}^k x_{ij}^k + \lambda_2 \sum_{i\in R} w_i\big(C_i + \gamma L_i\big) + \lambda_3 \sum_{k\in K} z_k \right), \tag{19}$$

where L_i are the lateness variables from (11). In this extended model the optimiser chooses not only routes and room assignments, but also how many teams to deploy, balancing finishing time, total traverse effort, risk-weighted clearance times, and the cost of activating additional (possibly high-tech) teams.

7.3 Discussion

The extensions in this section illustrate how the base sweep model can accommodate more realistic features while preserving a clear mathematical structure. Time-dependent hazard penalties via lateness

variables L_i and soft deadlines D_i allow the user to encode operational time targets for critical rooms, without sacrificing linearity. Scenario-based travel times t_{ij}^{ℓ} capture evolving smoke patterns through repeated solves of the same MILP, highlighting routes that are robust across plausible hazard evolutions.

On the technology side, room-level sensors can be represented entirely through modified sweep times and risk weights, and heterogeneous teams with different speeds and search capabilities can be incorporated through team-specific parameters t_{ij}^{k} and s_i^k. When desired, deployment variables z_k and the cost term $\lambda_3 \sum_k z_k$ allow the same optimisation framework to support high-level planning questions such as "how many teams and which specialised equipment should be deployed to this building?" These extensions suggest that the proposed model can serve as a flexible backbone for evaluating both operational tactics and longer-term investment in sensing and responder technology.

8 Conclusions

This paper presents a unified, graph-based optimization framework for planning building sweep operations under emergency conditions. By converting each floor plan into a node–edge network and formulating the sweep as a mixed-integer programme, our model integrates three essential operational objectives: minimizing total completion time, reducing unnecessary traversal, and prioritizing the clearance of high-risk rooms through a structured four-component weight system. The resulting optimization routes are intuitive, implementable, and consistent with real responder practices.

Across multiple building geometries—a one-story office, a five-floor school, and a single-family house—the model produces efficient and well-balanced sweep plans. Experiment 1 shows that our MILP significantly outperforms random, greedy, and manual-split strategies, particularly when responder resources are limited. These results highlight the model's ability to coordinate movement across floors, avoid redundant stair usage, and distribute workloads effectively.

Sensitivity analysis further demonstrates the value of the risk-weighted objective. Even moderate weights shift clearance priority toward vulnerable or hazard-prone rooms while keeping total sweep time nearly unchanged. Stronger weights push critical rooms even earlier in the sweep, offering a tunable trade-off between global efficiency and safety-focused protection. These behaviours confirm that the weighting mechanism is not cosmetic; it directly shapes meaningful operational decisions.

Finally, the model's structure naturally accommodates extensions, including soft deadlines for time-sensitive rooms, scenario-based smoke conditions, and technology-enhanced teams. These additions preserve linearity, ensuring that the framework remains computationally practical while capturing richer emergency dynamics.

Overall, the proposed model provides a flexible, physically interpretable, and quantitatively validated tool for responder routing. It supports both strategic planning and real-time decision-making, offering a foundation that can be adapted to a wide range of buildings and emergency scenarios.

9 Reference

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10 Letter to the Local Emergency Planning Committee

Dear Members of the Local Emergency Planning Committee,

We are writing to share our evacuation sweep strategies developed through a mathematical model applied to various building layouts: a one-story office, a five-floor school, and a single-family house. By converting each floor plan into a network of nodes and edges, we optimized responder routes to ensure rapid and safe sweeps. Our model uses a *priority weight* system, considering room function, occupant vulnerability, proximity to hazards, and lingering probability. For example, rooms such as nurseries and laboratories are prioritized higher than storage areas, ensuring that responders address the most vulnerable spaces first.

The following principles have been found to ensure efficient sweeps across different layouts. First, we recommend node-based planning, where building floor plans are digitized into graphs, enabling pre-assigned sweep routes that minimize time and maximize coverage. Second, we suggest assigning teams to adjacent zones rather than isolated rooms. In smaller layouts, such as a one-story office, two responders should start at the same entrance, each clear all rooms on either sides of the corridor, while in multi-floor buildings, responders clear zones floor by floor.

Third, priority-based routing is key—rooms with higher risks, such as those near kitchens, storage, or labs, should be cleared first. Vulnerable rooms, such as nurseries and classrooms, should also be prioritized. Redundancy is built into the plan, where high-priority rooms are checked by more than one team if possible, minimizing the chance of oversight. Furthermore, we recommend maintaining optimal team sizes: three responders work best for multi-floor schools, and adding extra responders yields minimal benefit beyond three teams.

For different building types, we suggest specific strategies. In office buildings, we recommend two teams per linear corridor, with a focus on visitor areas and rooms near hazard sources. Redundancy can be achieved by passing through cleared rooms to ensure no one is missed. For schools and childcare centers, we recommend three teams: one for lower floors, one for the central floor, and one for upper floors. High-priority rooms, such as classrooms near labs and kitchens, should be cleared first. For residential buildings, particularly complex homes with basements, three teams are ideal: one for the bedroom wing, one for the living room/basement, and one for the kitchen/garage. Critical areas, like nurseries and bedrooms, should always be first-priority zones.

In addition to these strategies, we suggest the use of modern technologies to improve sweep efficiency. Room sensors for smoke, CO, and occupancy in high-risk rooms can help reduce redundancy and speed up checks. Digital floor plans should be used to store pre-planned routes for quick reference during an emergency. Finally, thermal cameras and indoor positioning systems should be used by specialized teams assigned to high-risk areas, such as basements or labs.

In conclusion, our model shows that optimal evacuation sweeps rely on clear zone assignments, prioritization of high-risk rooms, and a manageable number of teams, with built-in redundancy. We hope these strategies, when adapted to local building inventories, will enhance emergency preparedness and evacuation efficiency.

Respectfully submitted,

Team 16486

Report on Use of AI Tools

Team Control Number: 16486
Problem: 2025 HiMCM Problem A – Emergency Evacuation Sweeps

During the preparation of our solution, we used three large language model (LLM) AI tools:

- **OpenAI ChatGPT** (GPT-5.1, via `chat.openai.com`, Oct–Nov 2025),
- **Anthropic Claude** (Claude 3 family, via `claude.ai`, Oct–Nov 2025),
- **Google Gemini** (Gemini 1.5 family, via Google AI interface, Oct–Nov 2025).

These tools were used only as *support tools* for writing, explanation, and coding assistance. All mathematical models, algorithms, numerical experiments, and final conclusions are our own work.

1. Types of Assistance

1.1 Report structure and section planning. We asked ChatGPT, Claude, and Gemini for suggestions on structuring the report within the 25-page limit, including where to place assumptions, model development, applications to different layouts, sensitivity analysis, and the letter to the local emergency planning committee. We compared the suggested outlines, then designed our own final structure and headings. The actual section order, titles, and transitions in the paper were chosen and edited by our team.

1.2 Language polishing and clarity of exposition. For paragraphs we had already drafted, we used all three tools to improve English clarity and concision *without changing technical meaning*. This included parts of the Introduction, model description, discussion of strengths and weaknesses, and the letter. Each tool proposed alternative phrasings; we treated these as drafts, kept only versions we agreed with, and often combined or edited them further. None of the tools introduced new modeling ideas, assumptions, or results; they only helped refine wording and style.

1.3 Explanation of our existing mathematical model. After we had independently formulated our mixed-integer linear program on a graph $G = (V, E)$, with decision variables for team routes, room assignments, arrival times, and finishing time, we asked ChatGPT, Claude, and Gemini to help us explain these equations more clearly in words. They suggested clearer descriptions for routing variables, flow-conservation constraints at nodes, big-M constraints for time consistency, and the objective function. We did not modify our sets, constraints, or objective based on AI output; the mathematical formulation remained exactly as designed and implemented by the team.

1.4 Code explanation and debugging guidance. We implemented our models in Python using a standard MILP solver. When we encountered errors (for example infeasible models, index issues, or suspiciously large big-M behavior), we provided small code snippets and error messages to one of the tools and asked for possible logical problems. Typical suggestions included checking balance constraints on intermediate nodes, verifying starting conditions for responders, and tightening big-M constants. We then manually revised our code, reran the solver, and validated correctness by confirming

that all required rooms were swept and that timing matched travel and sweep times according to our own checks.

1.5 LaTeX/TikZ and figure formatting. For figures such as floor plans, node graphs, and a high-level workflow diagram, we requested example LaTeX/TikZ templates from ChatGPT, Claude, or Gemini. We adjusted node positions, labels, fonts, and captions ourselves to match our actual building layouts and modeling logic. The final appearance and content of all figures reflect our own design; AI-generated code served only as a starting point.

2. Verification and Limits of AI Use

In all cases, we treated AI output as *advisory only*, not as authoritative answers:

- We checked all mathematical explanations, code hints, and polished text for accuracy, internal consistency, and alignment with the HiMCM problem statement.
- We rejected or corrected any suggestions that were unclear, unnecessary, or incompatible with our modeling decisions.
- We did **not** use AI tools to invent the core model, choose key parameters, generate numerical results, or interpret our findings. All modeling decisions, computations, and conclusions are our own.
- We are aware that LLMs may fabricate references or facts. Therefore, all references and citations in our paper were added, cross-checked, and formatted by team members, not taken directly from AI outputs.

No other AI tools (such as additional LLMs, code copilots, or automatic translators) were used. If further tools are used in a later revision, they will be documented in an updated AI report.